

Galileo's falling parabola, Newton's refinement and the Planck-scale question

For horizontal inertial speed v and downward force F , the area between the inertial horizontal line and the falling parabola obeys

$$\boxed{\frac{3F}{v}A_{\text{inertial,fall}} = \tau\Delta E, \quad \Delta E = \frac{F^2\tau^2}{2m}.}$$

This is the programme's geometric anchor. The physical question is what prevents arbitrarily fine Newtonian refinement from retaining distinguishable classical alternatives as this action-scaled area shrinks. Planck's constant sets quantum phase resolution; a hard prohibition on every smaller area or deterministic energy–time product requires an additional argument.

N01, 2026-09-14. The geometric coefficients reuse C001/C027 and the retained Principia derivation. Sections 3–4 add an exploratory closed comparison of inertial and piecewise falling motion, with written review and no ledger promotion. No novelty is asserted.

1. Which Newton area is being compared

Take $m, F, v, \tau > 0$ and downward coordinate y . The initial horizontal inertial trajectory and the falling trajectory are

$$q_I(t) = (vt, 0), \quad q_F(t) = \left(vt, \frac{Ft^2}{2m}\right), \quad 0 \leq t \leq \tau.$$

Close their geometric region by the vertical segment at $x = v\tau$. Its area and the gained vertical kinetic energy are

$$A_{\text{inertial,fall}} = \int_0^{v\tau} \frac{Fx^2}{2mv^2} dx = \frac{vF\tau^3}{6m}, \quad \Delta E = F\Delta y = \frac{F^2\tau^2}{2m}.$$

For gravity put $F = mg$. Thus the geometrical area is proportional to $\tau\Delta E$, with the indispensable factor $3F/v$. Its units are area; $\tau\Delta E$ has action units. This is the parabola and limiting-area geometry of the projectile Scholium and Lemmas X–XI in the retained Principia note. Kepler's equal-area law and sectors swept about a force centre are a separate construction.

Three elementary regions occur in this diagram:

Region	Area
Inertial horizontal line to falling parabola	$vF\tau^3/(6m)$
Matched-endpoint chord to falling parabola	$vF\tau^3/(12m)$
Inertial horizontal line to endpoint chord	$vF\tau^3/(4m)$

The first area is twice the second; the third is their sum. The programme's anchor is the first. The second is useful for comparing actions with the same endpoints, but must not silently replace it.

For $\mathcal{L} = m|\dot{q}|^2/2 + Fy$, the existing C001 calculation gives

$$D_{\text{chord,fall}} = S[q_{\text{chord}}] - S[q_F] = \frac{F^2\tau^3}{24m} = \frac{\tau\Delta E}{12} = \frac{F}{4v}A_{\text{inertial,fall}}.$$

Both paths in this last comparison have the same endpoint positions and times, so adding a total time derivative to the Lagrangian cancels from their action difference. The inertial horizontal path instead ends at a different height. Its action comparison needs endpoint/control phases before it is a measurable relative phase. Section 3 supplies an explicit reunion rather than hiding that requirement in the area-to-action coefficient.

2. The precise unresolved Planck step

At fixed m, F, v , these areas and action products scale as τ^3 . Inserting arbitrarily small mathematical intervals makes them approach zero. The research objective is a **physical obstruction to continuing that refinement as distinguishable classical motion**, or a stronger admissibility principle if one can be derived. Repeating the classical zero limit does not complete this objective.

The deterministic gain ΔE above is not a quantum state's energy standard deviation. Total mechanical energy is conserved during free fall; the kinetic gain is compensated by potential loss. Therefore a statement about energy uncertainty and an observable evolution time cannot be applied by simply renaming its two variables ΔE and τ .

With the quantum rule $e^{iS/\hbar}$, small action differences give nearby phases. Feynman's least-action discussion explicitly sums contributions from neighbouring paths, including paths with action differences within $\hbar$. It does not impose a smallest allowed path-action difference. This supports the breakdown of a uniquely resolved classical alternative, rather than an automatic cutoff on the continuum of paths. A theorem forbidding physical area shrinkage is a stronger target.

3. Close the inertial and falling paths with specified forces

Consider a horizontal reference arm with zero transverse force and a signal arm with the following controlled transverse force. Set $a = F/m$:

$$f(t) = \begin{cases} F, & 0 < t < \tau, \\ -F, & \tau < t < 3\tau, \\ F, & 3\tau < t < 4\tau. \end{cases}$$

Both arms start at $y = 0$ with $\dot{y} = 0$. The signal arm's first interval is exactly the falling parabola of Section 1. The remaining intervals return it to the reference arm with the same position and velocity. This is an ideal branch-controlled force experiment; a gravity realization needs compensating forces on the reference and return stages. It is not unassisted free fall throughout the whole comparison.

Let $P(t) = \int_0^t f(s) ds$ and $Y(t) = m^{-1} \int_0^t P(s) ds$. Then $P(4\tau) = Y(4\tau) = 0$. Explicitly P/F is t on the first interval, $2\tau - t$ on the second, and $t - 4\tau$ on the third. Consequently

$$\int_0^{4\tau} P(t)^2 dt = \frac{4}{3} F^2 \tau^3, \quad \int_0^{4\tau} Y(t) dt = \frac{2F\tau^3}{m}.$$

Y stays nonnegative and reaches $F\tau^2/m$ at $t = 2\tau$. At common horizontal speed v the area enclosed between the signal trajectory and its inertial baseline is

$$A_{\text{loop}} = v \int_0^{4\tau} Y(t) dt = \frac{2vF\tau^3}{m} = 12A_{\text{inertial,fall}}.$$

The transverse signal Lagrangian is $m\dot{Y}^2/2 + f(t)Y$; the reference transverse Lagrangian vanishes. Since $m\ddot{Y} = f$ and the endpoint boundary term $mY\dot{Y}$ vanishes,

$$\Delta S_{\text{loop}} = -\frac{1}{2m} \int_0^{4\tau} P(t)^2 dt = -\frac{2F^2\tau^3}{3m} = -\frac{F}{3v} A_{\text{loop}} = -\frac{4}{3}\tau\Delta E.$$

The sign and coefficient differ from the off-shell chord comparison, because the return forces are part of this experiment. The exact relation connects the original falling-parabola area to a closed, dynamically specified phase comparison: $|\Delta S_{\text{loop}}| = 4FA_{\text{inertial,fall}}/v$.

4. The same closed comparison in quantum mechanics

Assume canonical quantum kinematics $[y, p] = i\hbar$, a coherent two-arm label, and conditional Hamiltonians

$$H_I = \frac{p^2}{2m}, \quad H_F(t) = \frac{p^2}{2m} - f(t)y.$$

These assumptions supply $\hbar$ and the phase rule; this is a test of the Planck-scale interpretation, not a derivation of quantum mechanics from classical geometry. No additional branch-dependent scalar potentials or uncontrolled arm-label phases are included in the model. Horizontal evolution is common and cancels in the relative propagator.

In the interaction picture of H_I , the evolution generator is $B(t) = if(t)(y + tp/m)/\hbar$. Its commutator is the scalar

$$[B(t), B(s)] = \frac{if(t)f(s)(t-s)}{m\hbar}.$$

All higher nested commutators vanish. The first integrated term vanishes because $\int f = \int tf = 0$. The second gives the exact relative unitary

$$U_I(4\tau)^\dagger U_F(4\tau) = \exp \left[-\frac{i}{2m\hbar} \int_0^{4\tau} P(t)^2 dt \right] I = e^{i\Delta S_{\text{loop}}/\hbar} I.$$

For the sign, integrate $\int f(t) \int_0^t f(s)(t-s) ds dt = -\int P(t)^2 dt$; $P(4\tau) = 0$ eliminates the boundary term. The algebra can first be done on Schwartz states; the resulting forced free-particle unitaries extend the identity to the motional Hilbert space. Thus the two wavepackets reunite exactly for arbitrary common initial motional preparation, not just their classical centres. A coherent balanced arm label then carries relative phase $\phi = \Delta S_{\text{loop}}/\hbar$.

Use the already reviewed C006 comparison between this phase state and the zero-phase state, with equal hypothesis priors. For n independent copies and any joint binary quantum measurement, the success probability is

$$p_{\text{opt}} = \frac{1}{2} \left(1 + \sqrt{1 - \cos^{2n}(\phi/2)} \right).$$

Within the first lobe $|\phi| \leq \pi$, a required success probability $p_* > 1/2$ gives the action threshold

$$d_{n,p_*} = 2\hbar \arccos \left([4p_*(1-p_*)]^{1/(2n)} \right), \quad \tau \geq \left(\frac{3md_{n,p_*}}{2F^2} \right)^{1/3}.$$

Equivalently, in this specified comparison the original single falling lens must have $A_{\text{inertial,fall}} \geq vd_{n,p_*}/(4F)$ for that success rate. At $n = 1, p_* = 3/4$, $d = \pi\hbar/3$. For fixed $p_* < 1$ it tends to zero as $n \rightarrow \infty$; perfect discrimination instead requires $d = \pi\hbar$ for every finite n . Phase periodicity forbids extending the inequality beyond the first lobe as a monotone criterion.

This is a positive, protocol-dependent obstruction to resolving the shrinking Galileo area at fixed resources. It is not an absolute prohibition on smaller loops: their quantum states remain defined and approach the zero-phase state. It also does not assert that this interference experiment optimizes every possible measurement of the force or trajectory.

5. Research consequence and next decision

The double-check recovers the intended chain: **inertial line versus falling parabola -> action-scaled area -> Planck-sensitive distinguishability**. It identifies an actual drift source in STATE's substitution of Kepler swept sectors and in README's stale audit queue.

These entry points are corrected. The arc–chord area remains a secondary matched-endpoint comparison with a factor of two; it must not replace the user’s inertial–parabola anchor.

The closed-force construction supplies a quantum benchmark. The user’s intended innovation is stronger: **a mathematical necessity argument** that excludes zero-action refinement without assuming quantum mechanics, an uncertainty relation, a positive action unit or a fixed measurement budget. The quantum calculation above therefore supports the interpretation but does not discharge that objective.

N02 must construct and decisively test one independently justified consistency principle for joint time/position refinement of the Galileo comparison. State the cut state, composition and limiting variables. Derive the zero-action contradiction, or give an exact countermodel satisfying the proposed principle. C027–C029 and the full-state classical cuts already show why unqualified appeals to refinement consistency do not suffice; a new principle must do more than repeat those requirements. This returns to I003’s originating logical-obstruction question with its existing corrections retained. No mathematical contradiction follows merely from the fact that nature is quantum.